



Graphics: The Way Ahead

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DESKTOP COMPUTING IS at a crossroads. Until now, processor architecture was the dominant force that shaped this industry, but it's giving away to graphics.

New-age operating systems such as Windows Vista and Apple Leopard are completely based on graphical eye-candy, and need some serious graphics hardware to run. HD-quality movies demand a well sorted out graphical subsystem to deliver crisp, jerk-free video. Computer games are seeing unprecedented levels of realism, and to play them in their full glory, powerful graphics solution are needed. And as we move ahead, the need for better graphics is imperative.

The industry is gearing to deliver solutions. NVIDIA and ATI (now AMD) have already taken steps in the right direction. Even Intel has laid down the design for their new GPU codenamed Larrabee, expected by the second half of next year. It will be interesting to see how Intel handles its second attempt at graphics cards—their earlier attempt, the i740, never really accomplished much. Personally, I think they have a tough battle ahead against the likes of NVIDIA and AMD (ATI). Moreover, with the Nehalem architecture in the pipeline, Intel really *needs* expertise in graphics.

Today, NVIDIA has the best graphics card for the discrete graphics market, and the new 8800GT has raised the bar for powerful but affordable graphics. Almost all new laptops today are equipped with NVIDIA solutions, and

I'm sure the future will see the green logo appearing on cell phones. In a GPU-centred world, NVIDIA stand to benefit the most.

What I'm waiting for most are the Nehalem and Bulldozer from Intel and AMD respectively. Intel has already declared Nehalem the successor to the Penryn line of processors. Boasting of CPU+GPU cores in one single package, it will be a killer product for a value-sensitive market such as ours. All we'll need to worry about soon is which processor to buy—just one package will both process and play.

AMD has announced its Bulldozer architecture—a modular approach towards combining CPU+GPU cores in a single package to deliver the best of both worlds for a particular segment of consumers. After acquiring ATI, AMD is set to release Bulldozer before Intel's Nehalem. Regardless, we consumers have the most to gain from it all.

Coming back to what I said earlier, the transition to a GPU-dominated world is certain, and the promises are revolutionary. Sadly, for us geeks, the wait is torturous.

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